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Titan Alpha Apex 16 – Mathematical Closure Blueprint

1. Lattice Node Definition

Let the FCC base lattice have unit cell size $a = 0.09$ m. Canonical nodes $N_{i,j,k}$ are indexed by integers i, j, k in 3D space.

$$N_{i,j,k} = (ia, ja, ka), \quad i, j, k \in \mathbb{Z}$$

Standing wave peak assignment:

$$\text{Peak nodes } P_{i,j,k} = N_{i,j,k} + \delta_z(i, j, k)$$

Troughs (L3 expansion zones):

$$T_{i,j,k} = N_{i,j,k} - \delta_z(i, j, k)$$

$$\text{where } \delta_z(i, j, k) = A \cdot \sin \left(2\pi f \frac{\sqrt{i^2 + j^2 + k^2}}{v} \right)$$

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2. Triple Helix Layering

Each helix $H_m, m \in 0, 1, 2$ is offset by 120° in XY plane:

$$H_m : \begin{cases} x_{i,j,k}^{(m)} = ia + r \cos \left(\theta + \frac{2\pi m}{3} \right) \\ y_{i,j,k}^{(m)} = ja + r \sin \left(\theta + \frac{2\pi m}{3} \right) \\ z_{i,j,k}^{(m)} = ka \cdot \phi \end{cases}$$

- r = helix radius (fraction of a)
- $\theta = 2\pi k/L$ with L = helix pitch
- $\phi = 1.618$ (Golden Ratio vertical spacing)

Constraint:

$$|H_m - H_n| \geq d_{\min}, \quad m \neq n$$

3. Quasicrystal Expansion Zones (L3)

- Trough nodes allow lateral expansion $\Delta x, \Delta y$ under standing wave resonance:

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Basic Crystal Expansion Zones (L)



- Trough nodes allow lateral expansion $\Delta x, \Delta y$ under standing wave resonance:

$$T_{i,j,k}^{(m)} = H_{i,j,k}^{(m)} + (\Delta x_{i,j,k}, \Delta y_{i,j,k}, 0)$$

- Expansion constrained by:

$$\sqrt{\Delta x_{i,j,k}^2 + \Delta y_{i,j,k}^2} \leq \epsilon \cdot a$$

- ϵ = allowed expansion fraction (~ 0.075 φ tolerance)

4. Resonant Node Displacement

Standing wave imposes vertical displacement along z:

$$\delta_z(i, j, k) = A \sin(2\pi ft - k_s s_{i,j,k})$$

- $s_{i,j,k}$ = distance along wave vector from Node 0
- $k_s = 2\pi/\lambda$ = wavenumber
- $A \leq 0.075 \varphi$ for tolerance

$$|P_{i,j,k} - P_{i',j',k'}| \geq d_{\min}, \quad \forall i \neq i', j \neq j', k \neq k'$$

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5. Golden Ratio Constraints

- Vertical spacing between layers:

$$z_{k+1} - z_k = \phi \cdot a$$

- Horizontal spacing preserves FCC ratio, multiplied by ϕ if scaling for quasicrystal.

6. Tolerance Inequalities

- Node deviation $\leq \varphi$ tolerance:

$$|\delta_x|, |\delta_y|, |\delta_z| \leq 0.075 \cdot a$$

- Helix separation $\geq$ minimum distance:

$$\min(|H_m - H_n|) \geq 0.05a$$

- Wave amplitude must satisfy lattice locking:

$$A \leq \min\left(\frac{d_{\min}}{2}, 0.075a\right)$$